Horizontal launches

antocciphysics | TEACHER KEY | CPA

Learning target: Find flight time from vertical motion and then horizontal range.

Suggested first pass: 1, 2, 3, 7. Add independent, mixed and stretch work from the bank as time and student evidence permit. Estimated times are planning estimates, not completion requirements.

01. Bring back an earlier idea (retrieve) *Review: Unit 2, Day 7; 3 course days later.*
3 N/kg toward the center; divide the surface value by 25.

02. Explain a worked example (worked)
Height 20 m vertical; range 6 m horizontal. Determine fall time, then multiply horizontal velocity by time.

03. Look and explain (model)
Vertical motion determines time; labels must distinguish h from range.

04. Build the solution (practice)
8 m; 12 m.

05. Use the idea (reason)
No, it doubles time and range because fall time scales with√height.

06. Choose a method (mixed) *Review: Unit 1, Day 20; 12 course days later.*
$a=8\ m/s^2$ right; $v_f=40$ m/s right. Constant net force and mass throughout.

07. Exit check (exit)
15 m horizontally from base.

08. STRETCH: Apply it another way (stretch)
5 m/s hits target; 2.5 m/s lands 5 m from base.

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01. Bring back an earlier idea (retrieve) *Review: Unit 2, Day 7; 3 course days later.*
Center distances 3 R and 5 R; altitudes h/R=2 and 4.

02. Explain a worked example (worked)
Height 20 m vertical; range 6 m horizontal. Determine fall time, then multiply horizontal velocity by time.

03. Represent and interpret (model)
Vertical motion determines time; labels must distinguish h from range.

04. Solve and justify (practice)
t=√(90/10)=3 s; range 45 m.

05. Analyze the reasoning (reason)
No, it doubles time and range because fall time scales with√height.

06. Choose a method (mixed) *Review: Unit 1, Day 20; 12 course days later.*
x interval [-66, -54] km; y interval [72, 88] km. Simplified geometry, gravity, Earth motion or missing observations limit the model. Bounds are not probabilities.

07. Exit check (exit)
t 2 s; range 16 m; vertical velocity 20 m/s down.

08. STRETCH: Apply it another way (stretch)
Time 1.5 s, height 11.25 m; vertical impact speed 15 m/s; total speed √369≈19.2 m/s.